

AI for Communication and Marketing

Lab Exam – part 2

Academic Year: 2025/2026

Course: B.Sc. in Artificial Intelligence

1. Business Scenario

You are a **Data Scientist** for a leading E-commerce platform. The Marketing department is concerned about customer retention. You have been provided with a dataset named "data.xlsx" containing customer profiles, satisfaction scores, and transaction summaries.

Your goal is to build a predictive model to identify customers at risk of churning and provide actionable marketing insights.

2. Dataset

The file "data.xlsx" includes features such as:

- CustomerID - Unique customer ID
- Churn - Churn Flag
- Tenure - Tenure of customer with the company
- PreferredLoginDevice - Preferred login device of customer
- CityTier - City tier
- WarehouseToHome - Distance in between warehouse to home of customer
- PreferredPaymentMode - Preferred payment method of customer
- Gender - Gender of customer
- HourSpendOnApp - Number of hours spent on mobile application or website
- NumberOfDeviceRegistered - Total number of devices is registered on particular customer
- PreferredOrderCat - Preferred order category of customer in last month
- SatisfactionScore - Satisfactory score of customer on service
- MaritalStatus - Marital status of customer
- NumberOfAddress - Total number of address
- Complain - Any complaint has been raised in last month
- OrderAmountHikeFromlastYear - Percentage increases in order from last year
- CouponUsed - Total number of coupon used last month
- OrderCount - Total number of orders placed last month
- DaySinceLastOrder - Day Since last order
- CashbackAmount - Average cashback in last month

3. Exam Requirements

Your submission must consist of two parts:

1. **A Jupyter Notebook (.ipynb) or standard Python .py scripts:** Containing clean, commented Python code
2. **A Presentation (max 3 Slides .pdf format):** Summarizing key findings and marketing recommendations for non-technical stakeholders

Task A: Data Audit & Quality

- Load the dataset and perform an initial audit: identify missing values and handle them appropriately (e.g., imputation or removal).
- Check for data consistency and outliers (e.g., check WarehouseToHome and Tenure ranges).
- Encode categorical variables for machine learning compatibility.
- Document your cleaning process: Why did you choose to drop, impute, or transform specific data points?

Task B: Exploratory Data Analysis (EDA)

- Analyze the correlation between Complain, SatisfactionScore and Churn.
- Visualize how Tenure and DaySinceLastOrder influence the probability of churn.
- Comment on the class balance: is the dataset skewed towards churners or non-churners?

Task C: Modeling & Evaluation

- Split the data into training and testing sets.
- Implement a Propensity Model using at least two algorithms (e.g., Logistic Regression vs. Random Forest or XGBoost).
- Evaluate performance using a Confusion Matrix, F1-Score and Lift. Pay special attention to Recall: why is catching a potential churning more important than missing a loyal one?

Task D: Presentation Deliverable (max 3 Slides)

- Risk Drivers. Identify the top 3 factors that drive customers away.
- Strategic Recommendation. Propose a specific marketing campaign (e.g., targeted coupons or customer service outreach) based on the model's results.